

UC Berkeley Math 128A: Homework 5

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MATLAB usage classification: PEN (pen and paper only), CALC (use as calculator), CODE (run codes).

0. Please assign yourself a self grade on this homework (0, 1, or 2) based off the guidelines in the syllabus.

1. (3.5-4c, CODE) Construct the natural cubic spline for the following data.

c.

x	$f(x)$
0.1	-0.29004996
0.2	-0.56079734
0.3	-0.81401972

2. (3.5-6c, CALC) The data in the previous exercise was generated using the following function. Use the cubic splines constructed for the given value of x for approximate $f(x)$ and $f'(x)$, and calculate the actual error.

c. $f(x) = x^2 \cos x - 3x$; approximate $f(0.18)$ and $f'(0.18)$.

3. (3.5-8c, CODE) Construct the clamped cubic spline using the data in the previous two exercises, and the fact that

c. $f'(0.1) = -2.8004996$ and $f'(0.3) = -2.453394925124157$

4. (4.1-6a, CALC) Use the most accurate three-point formula to determine each missing entry in the following tables.

a.

x	$f(x)$	$f'(x)$
-0.3	-0.27652	
-0.2	-0.25074	
-0.1	-0.16134	
0	0	

5. (4.1-8a, CALC) The data in the previous exercise was taken from the following function. Compute the actual error in the approximation, and find error bounds using the error formulas.

a. $f(x) = e^{2x} - \cos 2x$

6. (4.1-24/22, PEN) Derive an $O(h^4)$ five-point formula to approximate $f'(x_0)$ that uses $f(x_0 - h)$, $f(x_0)$, $f(x_0 + h)$, $f(x_0 + 2h)$, and $f(x_0 + 3h)$. [Hint: Consider the expression $Af(x_0 - h) + Bf(x_0 + h) + Cf(x_0 + 2h) + Df(x_0 + 3h)$. Expand in fourth Taylor polynomials, and choose A, B, C , and D appropriately.]

7. (4.1-29, PEN) Consider the function

$$e(h) = \frac{\varepsilon}{h} + \frac{h^2}{6}M$$

where M is a bound for the third derivative of a function. Show that $e(h)$ has a minimum at $\sqrt[3]{3\varepsilon/M}$.

8. (4.2-1d, CALC) Apply Richardson's Extrapolation on the centered-difference formula:

$$f'(x_0) = \frac{1}{2h} [f(x_0 + h) - f(x_0 - h)] - \frac{h^2}{6} f'''(x_0) - \frac{h^4}{120} f^{(5)}(x_0) - \dots$$

to determine $N_3(h)$, an approximation to $f'(x_0)$, for the following function and stepsize.

d. $f(x) = x^3 \cos x$, $x_0 = 2.3$, $h = 0.4$

9. (4.2-2d, CALC) Add another line to the extrapolation table in the previous exercise to obtain the approximation $N_4(h)$.
10. (4.2-8, PEN) The forward-difference formula can be expressed as

$$f'(x_0) = \frac{1}{h} [f(x_0 + h) - f(x_0)] - \frac{h}{2} f''(x_0) - \frac{h^2}{6} f'''(x_0) + O(h^3)$$

Use extrapolation to derive an $O(h^3)$ formula for $f'(x_0)$